

Fiona C. Dubay

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Education

Sarah Lawrence College

B.A. in Liberal Arts | *Cumulative GPA 3.93*

Concentration: Environmental Science

Academic Advisor: Drew Cressman | **Research Advisor:** Bernice Rosenzweig

Bronxville, NY

May 2024

Research Experience

NYC's Climate Vulnerability, Impact, and Adaptation Analysis (VIA)

New York City, NY

Student Intern

September 2023 - December 2023

- Created digital maps and visualizations using QGIS software for publication on flood exposure and risk in New York City
- Processed spatial data and databases in vector and raster formats for clear visualization in QGIS
- Communicated and collaborated with other scientists and specialists on the task team
- Assessed the quality of 311 data for analysis and visualization usage

NYC's Panel on Climate Change (NPCC) & VIA

New York City, NY

Summer Fellow

June 2023 - August 2023

- Crafted visualizations on flood exposure in NYC for publication in the NPCC's first chapter, specifically focused on flooding
- Sorted and cleaned open-source data from NYC OpenData's site for analysis use
- Handled and refined large, detailed, spatial data files through QGIS
- Utilized Python for cleaning, refining, and visualizing rainfall and tide gauge datasets

Sarah Lawrence College Summer Research Program

New York City, NY

Environmental Science Intern

2021, 2022

Advisor: Bernice Rosenzweig

- Engaged in hands-on research experience through an intensive 8-week summer program
- Assisted faculty with existing research work
- Gained a robust understanding of the means of conducting research through individual and collaborative projects
- Designed and carried out experiences, analyzed results, communicated findings with a wide audience through presentations

Sarah Lawrence College Conference Work

Bronxville, NY

10+ research projects completed complementary to coursework,

2020 - 2024

on topics ranging from hydrology, remote sensing, microbiology, and ecology.

Link to research portfolio containing these projects, and more: fionadubay.com

Examples:

- "Case Study: Vinyl Chloride Train Derailment in East Palestine, OH"
- "Terrain Analysis and Urban Watersheds: Delineating the Upper Bronx River Basin"
- "Habituation and other complex processes in *Physarum polycephalum*: what a single-celled organism can teach us about how we learn"

Honors & Awards

Sarah Lawrence College Edward Cogan Prize for Science and Mathematics	2024
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Sarah Lawrence College Scholarships:	
Gaylord Donnelley Family Science Scholarship	2020-2024
Dean's Scholarship	2020-2024

Research Interests

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- Fate, transport, and remediation of environmental contaminants
 - Hydrology & hydrological modeling
 - Applications of remote sensing and environmental modeling
 - Urban ecology

Work Experience

City of Santa Monica Office of Sustainability & the Environment	<i>Santa Monica, CA</i>
<i>CivicSpark / AmeriCorps Fellow</i>	October 2025 - Present
• Assist with community planning projects, such as developing an extreme heat risk action plan • Mobilize volunteers and gather community feedback to support project implementation • Gather, analyze, and organize data for various analyses and visualizations that support ongoing projects	

CivilGrid	<i>Remote</i>
<i>Data Research Specialist</i>	October 2024 - February 2025
• Interpreted maps, as-built drawings, and engineering plans • Researched utility operations, public works assets, and asset management plans • Digitized utility maps in ArcGIS Pro • Communicated directly with public works offices and utility asset owners	

Agerpoint	<i>Remote</i>
<i>Geospatial Data Intern</i>	May 2024 - August 2024
• Ingested and processed orthomosaic and LiDAR data based on Standard Operating Procedures (SOPs) • Assisted in machine learning model training by providing feedback to machine learning/AI development teams on model performance • Organized and maintained data storage according to client account requirements and standards	

Technical Skills

Geospatial Analysis: QGIS, ArcGIS Pro
Data Analysis: Python (NumPy, Pandas, geopandas), R
Visualization: Python (matplotlib, seaborn), QGIS, ArcGIS Pro
Additional: CPR & First Aid Certification

Volunteer Experience

MESD Outdoor School	2017-2019
Student Leader / Environmental Educator	600 hours